

Basic Reminders

Collection of some relevant basic notions

$O(\cdot)$ and $o(\cdot)$

If $f, g : \mathbb{R} \rightarrow \mathbb{R}_+^*$, then

$$f(x) = O(g(x)) \text{ means } \exists x_0 \in \mathbb{R} \exists C \in \mathbb{R} \geq 0 : \frac{f(x)}{g(x)} \leq C \text{ for } x > x_0$$

That is, $f(n)$ is *eventually* bounded by $g(x) \times$ a constant. One basic example is $f \in O(1)$, which means f is eventually bounded. If $f, g : \mathbb{N} \rightarrow \mathbb{R}_+$, then eventually bounded and bounded coincide. For sequences, one relevant example is a statement of the form $f \in O(1/n)$. This means that $f(n)$ is at most $\frac{1}{n} \times$ a constant – by the squeeze law, this means that $f(n) \rightarrow 0$ at least as fast as $1/n$. Put another way, if $f(n)$ represents an *error* at iteration n , then $f \in O(1/n)$ means that the error is *at most* $1/n \times$ a constant.

There are a few basic rules about the classes of $O(g(n))$ functions/sequences that are important to recall:

1. $f \in O(g(n))$ and $g(n) \leq h(n) \Rightarrow f \in O(h(n))$.
2. $\alpha \in \mathbb{R}$ and $f \in O(g(n)) \Rightarrow \alpha f \in O(g(n))$. (α gets absorbed into the $O(\cdot)$)
3. $f, g \in O(h(n)) \Rightarrow f + g \in O(h(n))$ ($\max(C_1, C_2)$)

If $f, g : \mathbb{R} \rightarrow \mathbb{R}_+^*$, then

$$f(x) = o(g(x)) \text{ means } \forall \varepsilon > 0 \exists x_0 \in \mathbb{R} : \frac{f(x)}{g(x)} < \varepsilon \text{ for } x > x_0$$

which is equivalent to saying that $f(x)/g(x) \rightarrow 0$ as $x \rightarrow \infty$. That is, if $g \rightarrow 0$ then $f \rightarrow 0$ *strictly faster* than $g(x) \rightarrow 0$. But it is not necessary that $g \rightarrow 0$, just that f is eventually negligible compared to g . Clearly, if $f \in o(g)$, then $f \in O(g)$.